

University of Essex Online

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Assignment: Development Individual Project: Presentation

Project title: Machine learning on paediatric pneumonia

X-rays dataset. HOG + SVM vs ResNet50

Transcript

Slide 1: Title

Hello and welcome. Today I will present to you my findings from the modeling and analysis in classical machine learning and deep learning for the X-ray, paediatric chest dataset. On one side, a classical model, HOG plus SVM, and on the other, a deep learning model, ResNet50.

Slide 2: Significance and motivation

The motivation is twofold. On the one hand, it's personal, as a kid, I had regular asthma, and was usually catching pneumonia. So this topic is of personal interest to me. On the other hand, and the more academic and technical for this assignment, pneumonia in kids is one of the major causes of death worldwide (UNICEF, 2024), unfortunately. Because the cost of a misdiagnosed pneumonia is way higher than a false alarm. So that motivates me to see how machine learning can contribute, and to close, technically the exercise represents a challenge due to the need to research, especially in deep learning and for this very particular use case.

Slide 3: Dataset

The dataset is publicly available in Kaggle as a redistribution (Mooney, 2018), coming from the Kermany et al. (2018) paper. One particularity that stands out is the image is paediatric, meaning there are no adults, and from a single hospital. So, therefore, the transferability is very limited. The class balance leans 3 to 1 towards pneumonia, which is the kind of imbalance that must be addressed. And from the EDA, I noticed that the image carries hints, like text labels or tubes, in the image itself. And that could fit into the typical shortcut learning risk that Zech et al. (2018) mentions. So I'm flagging this for later when we look at the explainability of the heatmaps.

Slide 4: Class imbalance and split

So first, the dataset is imbalanced, as mentioned before, I used balanced class weights in the SVM and a class-weighted loss in the CNN (King and Zeng, 2001). Plus, early stopping on the CNN validation AUC (Prechelt, 1998), to reduce overfitting.

Second, the supplied validation folder contained only 16 images, which is way too small to make any reliable choice about the models. As an alternative, by combining those with the training data, I made a new stratified 80/20 development split, keeping classes in the same proportion, following a standard model assessment practice (James et al., 2023). The official test set of 624 images was kept untouched for the final evaluation.

On top of that, I also ran five-fold stratified cross-validation (Kohavi, 1995), five folds gave me a stronger check than relying on a single split. The SVM is very stable across folds. The CNN on a reduced training schedule to keep compute reasonable is also stable, but with a wider spread. And I report macro-F1 (Sokolova and Lapalme, 2009) and ROC-AUC (Fawcett, 2006), alongside the accuracy. The latter being potentially misleading when classes like in this case are imbalanced (Saito and Rehmsmeier, 2015).

Slide 5: Preprocessing workflows

So, the dataset needs to be prepared, normalized, in order to accommodate each model needs. And as the model differences required it, so thus I ended up preparing 2 separate workflows for it. For SVM, the images are converted from color to grayscale, and resized, and scaled to sit between zero and one, so HOG and the downstream scaler, work on a consistent range. And then I extracted HOG features following Dalal and Triggs (2005).

For the CNN, it's based on the standard ResNet50 input format that the base was pre-trained on (He et al., 2016). With light data augmentation (Shorten and Khoshgoftaar, 2019), like rotations, shifts, zooms, and horizontal flips. Both workflows are fed into the same training, validation, and test splits. So the comparison is fair and equal.

Slide 6: HOG + SVM design

On the design for the classical machine learning, modeling, I went with HOG features fed into a support vector machine with an RBF kernel (Cortes and Vapnik, 1995). HOG is there because it gives a structured edge and shape representation that is more suitable for this type of baseline image. I also added Platt scaling (Platt, 1999) to give the SVM probability like outputs for the demonstration. The threshold analysis and the probability-based occlusion heatmaps will come later. I tuned with the standard scikit-learn defaults, for the RBF kernel and balanced class weights, for the class balance, and the classical pipeline was implemented in scikit-learn (Pedregosa et al., 2011).

About alternative classical models, I did consider others, such as linear SVM, random forest, and gradient boosting. But I did not run a full benchmark on all of them. So I ended up selecting the RBF SVM as an established non-linear baseline, following the literature that I followed (Hsu, Chang and Lin, 2003) and kept the exercise focused on the classical versus the deep learning comparison.

Slide 7: ResNet50 design

On the deep learning side, I used ResNet50 with transfer learning, which means it's a pre-trained model already trained on ImageNet, that has learned a lot about images. So thus I don't have to train one from scratch, which is the standard recipe for medical imaging when the data is limited (Cheplygina, de Bruijne and Pluim, 2019), like in this case. In practice, ResNet50 is a 50-layer convolutional neural network introduced by He et al. (2016). And the key innovation is the so-called residual connections that allow training of very deep networks that before were impossible to train properly.

The approach I follow is transfer learning (Yosinski et al., 2014), so the base of the network stays frozen and only the head learns from my dataset. This is the standard recipe described in the Keras Transfer Learning Guide (TensorFlow, 2024). The head itself is just a small classifier with a dropout layer to reduce overfitting (Srivastava et al., 2014), and a final sigmoid that outputs the pneumonia probability in this case. For training, I used Adam as the optimizer (Kingma and Ba, 2015) and early stopping on the validation AUC (Prechelt, 1998). So therefore, the model stops before it starts to overfit.

Why ResNet50 and not something else? Two alternatives were considered, the 1st one was training a CNN from scratch instead of using transfer learning. The 2nd alternative was DenseNet121, but for this assignment, I chose ResNet50 for reproducibility and compute practicality purposes.

Slide 8: Test-set comparison

Okay, so now the moment of truth. Let's review the results. Both models were tested on the same held-out test set, the official one that comes with the dataset. So the results tell us that the 4 key metrics point in the same direction, CNN seems to be more accurate than SVM, more specifically accuracy is 0.90 versus 0.76. Macro precision 0.90 versus 0.85 and macro recall 0.89 versus 0.68. On macro-F1, the headline metric on imbalanced data (Sokolova and Lapalme, 2009), the gap is 21 points. 0.89 for the CNN against 0.68 for the SVM. The highlight is on NORMAL recall, where SVM predicts only 36% of cases under the NORMAL category, whereas the CNN, 82%. So the conclusion is SVM heavily over-predicts pneumonia. So these numbers enable a cautious path to say CNN could be a good tool for assisted detection of pneumonia under these settings and design.

Slide 9: Confusion matrices

So now let's look at the confusion matrices for both models, which is basically the table that shows what the model got right and what the model got wrong. The bottom left cell in each matrix is the worst error. That is missed pneumonia because clinically, that's the kind of mistake that has the bigger consequence. This is the cost-sensitive classification framing from Provost and Fawcett (2001).

Looking at the SVM, of the 234 NORMAL X-rays, it gets 85 right. However, 149 are flagged as pneumonia. Basically, false alarms. On the pneumonia side, it gets almost all right, 388 out of 390, only missing 2 cases. So the SVM heavily, as we said before, over-predicts pneumonia. The CNN looks more balanced, 193 of 234 NORMAL are correctly identified with 41 false alarms. And 369 of 390 pneumonia cases are caught with 21 missed. So we have 2 very different error profiles. But in this case, these results raise some question marks because the SVM is almost not missing any pneumonia case, but at the cost of tons of false alarms. So we need to decide what is better.

Slide 10: Threshold trade-off

So the confusion matrices that we saw before, were with the 0.5 default, that comes from the model. So now, I generated these curves to understand what happens when we move the threshold, below or above the 0.5, right? So we see that the curve suggests that we could explore moving out of the 0.5 to approximately between 0.05 and 0.3. Why? Because in that range, the model catches more pneumonia cases, while still keeping the false alarms manageable. So let's remember what is the cost of a false alarm versus what is the cost of missing a pneumonia case? So the final cutoff for the definition of the threshold should be selected together with the doctors, with the clinicians. This is not a technical decision that machine learning can provide, but rather we can provide with the data. I will come back to this recommendation later when I show what happens when both models are tuned to catch the same percentage of pneumonia. And this is where the CNN looks like the better candidate.

Slide 11: Explainability - SVM

Okay, so we have shown how the models perform with numbers and confusion matrices, but how do we know the model is looking at the right thing on the image? I chose occlusion sensitivity (Zeiler and Fergus, 2014) for 3 reasons. First, it's model-agnostic, second, it doesn't require access to model gradients, which we don't have, and third, the results are represented by a heatmap, of which regions of the image matter the most for the decision. So all are compatible with the models and use case we are working on.

The method itself is very simple, cover a small patch of the X-ray, ask the model to predict again, and see how the probability changes. We do this for every patch and we end up with the heatmap.

The slide shows 2 SVM cases. The correctly classified pneumonia is where the SVM is very confident, almost 100%. So covering patches barely changes anything. The heatmap looks mostly empty. However, the misclassified normal is more useful. The heatmap shows score sensitivity around the shoulders and the edges of the image, not the lung field. This is a warning sign that the SVM could be using shortcut features to

decide, like text labels or scanner artefacts (Zech et al., 2018). On the next slide, we see the same method on the CNN, so we can compare both.

Slide 12: Explainability - CNN

So now for the CNN, I applied the same occlusion technique so the comparison is fair. The correctly classified pneumonia is where the CNN was extremely confident, above 99%, so just like the SVM, covering patches barely changes the probability. The misclassified normal is more useful again. The heatmap shows that the CNN response drifts towards the shoulders and the edges of the image. Not where the lungs are. So, even with a more sophisticated model, the same warning sign about shortcut features appears. One important caveat from the literature. Saliency-style maps can look meaningful without being faithful (Adebayo et al., 2018). So I treat the warm regions as a directional hint, not a causal explanation. Occlusion sensitivity is very computationally heavy. So a faster gradient-based method, such as Grad-CAM (Selvaraju et al., 2017), is suggested to use for production deployments. The shortcut warning on both models is one of the strong reasons for post-hoc investigation, which falls outside the scope and time of the current project, but very worth noting.

Slide 13: Demonstration

Okay, and now we will do a small demo, where both models will run on an image they have not seen before. So this is to demonstrate both models are working as per assignment requirement. As you can see, this is a NORMAL chest X-ray. I'm highlighting the results. The SVM predicts pneumonia with 0.83 probability, so it's wrong on this one. CNN gets 0.05 and correctly says NORMAL. This is the over-prediction problem from the previous slides, made concrete on one real case.

Slide 14: Subgroup performance

In statistics, we know that averages sometimes hide the specifics in the structure, and that's the subgroup analysis we will do now. In this case, I split the dataset into subgroups. These are subtype bacterial or viral pneumonia, the brightness of each image low mid and high, and the image size, small, medium, large. And the results are that the subtype split, the macro-F1 looks low for both bacterial and viral pneumonia, but that's a metric artefact. Only pneumonia in those groups. So we read PNEUMONIA recall around 0.99 for the SVM and 0.95 for the CNN. A smaller gap than the headline. Those subtype labels come from filenames, so we are not sure how confident we can be in this category. On the brightness one, the SVM drops as the image gets brighter, 0.74 down to 0.63. While the CNN holds steady, 0.85 to 0.94. So the CNN is more stable across lighting. On image size, the CNN PNEUMONIA recall drops from 1.0 on small to

0.75 on large, meaning almost a quarter of pneumonia cases on large images are missed. This subgroup check follows the fairness analysis pattern in Buolamwini and Gebru (2018) and Obermeyer et al. (2019), applied here to the specific characteristics of the image.

Slide 15: Ethics

So on the ethics. First is privacy. Although in this dataset, the data seems to be cleaned out with no apparent visible personally identifiable information in the files, there are still some visible artefacts. I'm unsure what exactly those are, but as a kind of warning, I will say go beyond metadata cleanup. Also, the images need to have a scan to avoid the name, date, or other sensitive information in the image itself.

Second, the error cost equation. A missed pneumonia in a kid can have serious consequences while a false alarm creates extra workload from a specialist to review. So the threshold we review that is by default at 0.5 is useful to gauge up or down, depending on how many false positives versus missed, pneumonia cases, we are willing to sustain? And the decision should not be taken by the engineering team, but by the clinical responsible team, the cost-sensitive classification argument from Provost and Fawcett (2001).

Third, the bias. The model only saw data from one hospital in Guangzhou. So Zech et al. (2018) showed that AI sometimes gets biased on specifics of the surrounding elements of the hospital that creates the dataset. Like, for example, scanner artefacts instead of the disease itself. So being this dataset exclusively from one hospital, I will treat it with caution. Out-of-distribution detection from Hendrycks and Gimpel (2017), could assist in mitigating this bias, but is only partial mitigation.

So, to close beyond the performance number and the model that performs the best, the conclusion is that machine learning definitely can be an assistance tool to reduce diagnosis times and costs. But not yet an automatic decision. Human oversight is highly needed and governance over the solution as well (Leslie, 2019).

Slide 16: Deployment architecture

So this is a conceptual implementation proposal, and what you can see is a component diagram of the different components of the X-ray pneumonia system predictor based on Azure (Microsoft, 2024). So on the left hand side, we have the hospital network. With the endpoints that take the chest X-rays at the hospital itself. And it sends them through DICOMweb, which is the standard protocol, to an Azure machine learning system on the cloud. Because this is a managed cloud service, it can scale up or down as the volume of images or users grows over time. So scalability is not a limitation. Obviously, there are costs involved. Only ResNet50 serves the pneumonia score. The SVM remains an

offline research comparator or for certain specific cases. The X-ray specialist sees the score and the associated heatmap, available so they can review and decide if they need to take a further look, or if the numbers are good enough to accept a conclusion. Remember the threshold at 0.5 that is the default. The X-ray specialist team can be trained to pay attention and be provided with metrics in a way that periodically they commit and decide if to gauge the threshold up or down based on the actual experience. As for machine learning operations, monitoring tracks the latency, drift, and subgroup performance, and clinician confirmations become labeled feedback for retraining (Sendak et al., 2020).

Slide 17: Summary and conclusions

Okay, so bringing everything together. Across the main measures, ResNet50 outperforms HOG plus SVM. The SVM does perform better on pneumonia recall at the 0.5 reporting cutoff, but it gets there by heavily over-predicting pneumonia. The CNN is more balanced, though it has the large image weakness. So my conclusion and recommendation is that ResNet50 is the model candidate based on the metrics.

One useful argument comes from the matched-sensitivity comparison at the same level of pneumonia detection around 99%. The CNN produces about 102 false alarms versus 149 for the SVM. I didn't separately perform an exercise to search for the best cutoff for each model, but this is an important pending task.

But the CNN has some warning signs too. We saw the subgroup drop to 0.75 recall on large images, and the apparent shortcut learning on the explainability heatmaps. So production deployment needs further evaluation on borderline cases (Sculley et al., 2015). Remember that the dataset and the process had its limitations too. For example, paediatric only on a single hospital. So transferability and further deployment are limited by these conditions (Zech et al., 2018).

Slide 18: Critical reflections and close

So to close a couple of lessons learned and some next steps that I take with me, beyond the metrics.

So as I said, this was a much more ambitious project than I expected. I started without much Keras knowledge and had to learn it the hard way. The most painful one was the Keras nested-model trainable flag. What I thought was a two-phase fine-tuning was actually not, because when the parent model is set to non-trainable, the inner layers stay frozen, even if you mark them as trainable. I only spotted this after re-reading the TensorFlow docs (TensorFlow, 2024), following troubleshooting. Big lesson learned about reading framework documentation carefully.

Another lesson was deep learning reproducibility. It used to be with classical machine learning, with a fixed seed for reproducibility, but CNN training on the same data drifts, a few F1 points between runs, because of the GPU being non-deterministic, dropout randomness and batch shuffling.

On the compute side, one notebook run took 9 hours on Colab, because the free-tier GPU fell back to CPU without telling me. Eventually, I had to upgrade to Colab Pro to rent an A100 just to run the CNN cross-validation in a reasonable time.

And finally, the domain learning. I came in without chest X-ray experience and quickly realized most of the design choices, pre-processing, subgroup splits, shortcut learning, ethics framings, and so on, were only possible if I understood the imaging and clinical context. So a big chunk of the work was learning the medical and image-processing side, not just the machine learning.

If I do the project again or have a follow-up, these are my priorities as next steps. Number one, rerunning the fine tuning after fixing the Keras trainable flag. Number two, external validation first on another paediatric hospital dataset and ideally include others too for an additional dimension. Number three, produce a DenseNet121 evaluation, as I was short on time to do that against the selected model.

All in all, this was an ambitious project for me. The technical complexity caught me by surprise more than once, but I learned much more than I expected at the start. Thank you.

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